Final Project: Credit Card Defaulter Classification Analysis

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1. Introduction

Credit card default prediction is a critical task in financial risk management, as the ability to accurately identify high-risk customers directly affects institutional profitability and stability. Financial institutions must strike a balance between minimizing financial losses caused by defaulters and avoiding the misclassification of reliable customers. This problem becomes particularly challenging when dealing with imbalanced datasets, where non-defaulters significantly outnumber defaulters.

This study investigates the effectiveness of advanced tree-based machine learning methods in predicting credit card default behavior. Specifically, Bagging, Random Forest, and Boosting are applied and compared against a baseline Lasso Logistic Regression model. The goal is to examine how model complexity, parameter tuning, and learning mechanisms influence predictive performance, particularly in detecting default cases. The analysis demonstrates that while simpler models offer interpretability, more flexible ensemble methods significantly improve the identification of defaulters and provide a better balance between different evaluation metrics.

2. Problem Definition and Model

2.1 Task Definition

The problem is formulated as a binary classification task in which the objective is to predict whether a credit card customer will default in the following month. The predictors include repayment status variables, billing amounts, payment amounts, and demographic characteristics such as age, education, marital status, and gender. The response variable indicates whether a customer defaults or not.

This problem is non-trivial because the dataset (Shashank Necrothapa. (n.d.). *Ames housing dataset* [Data set]. Kaggle.) exhibits class imbalance, making it difficult for models to accurately detect the minority class. In addition, the relationships between predictors and the outcome are likely nonlinear and involve complex interactions, particularly in financial behavior variables. These characteristics justify the use of advanced modeling techniques beyond simple linear approaches.

2.2 Model Definition

This study employs four different modeling approaches, including Lasso Logistic Regression as a baseline and three tree-based ensemble methods: Bagging, Random Forest, and Boosting. Each model follows a different learning mechanism, which is described below in detail.

The Lasso Logistic Regression model is a linear classification method that estimates the probability of default using a logistic function. Given input features, the model computes a linear combination of predictors and applies a sigmoid transformation to produce probabilities. Lasso introduces an L1 regularization term to the loss function, which shrinks some coefficients toward zero. As a result, it performs automatic feature selection by retaining only the most relevant predictors.

The Bagging method constructs multiple decision trees using bootstrap sampling. For each tree, a random sample of the training data is drawn with replacement, and a full decision tree is grown without pruning. This process is repeated for a predefined number of trees. During prediction, each tree

independently produces a class label, and the final prediction is obtained by majority voting across all trees. The overall process can be summarized as follows: first, generate multiple bootstrap samples from the training data; second, train a decision tree on each sample; third, aggregate predictions from all trees using majority voting.

Random Forest extends the Bagging approach by introducing additional randomness during tree construction. In addition to bootstrap sampling, only a random subset of predictors is considered at each split in the tree. This prevents the model from relying too heavily on dominant predictors and reduces correlation among trees. The algorithm proceeds by repeatedly selecting a random subset of features at each split, identifying the best split among those features, and growing trees fully. Predictions are obtained by averaging or voting across all trees. The key parameter in this model is the number of predictors considered at each split, denoted as mtry.

Boosting differs fundamentally from Bagging and Random Forest in that it builds trees sequentially rather than independently. The model starts with an initial prediction and iteratively adds trees that focus on correcting the errors of previous iterations. At each step, a new tree is fitted to the residuals or misclassified observations from the previous model. The contribution of each tree is controlled by a learning rate parameter known as shrinkage. The final prediction is obtained by summing the contributions of all trees. The algorithm can be described as follows: initialize the model with a constant prediction; for each iteration, compute the residual errors; fit a small decision tree to the residuals; update the model by adding the scaled predictions of the new tree; repeat until the specified number of trees is reached.

These models provide a structured comparison between linear and nonlinear approaches. While Lasso relies on linear relationships and regularization, the tree-based methods capture nonlinear patterns and interactions, making them more suitable for complex classification problems such as credit default prediction.

The choice of models in this study is guided by the nature of the problem and the characteristics of the dataset. Credit default prediction involves complex and potentially nonlinear relationships between financial behavior variables and the outcome. As a result, relying on a single modeling approach may not capture the full structure of the data.

Lasso Logistic Regression is selected as the baseline model because it provides a simple and interpretable framework for classification while performing automatic feature selection through regularization. It serves as a benchmark to evaluate whether more complex models provide meaningful improvements.

Bagging is chosen to reduce variance by averaging multiple decision trees trained on different bootstrap samples. Since decision trees are highly sensitive to small changes in data, Bagging improves stability and reduces overfitting without altering the underlying tree structure.

Random Forest extends Bagging by introducing randomness in feature selection at each split. This is particularly useful in datasets where certain predictors, such as repayment variables, may dominate the model. By forcing the model to consider different subsets of variables, Random Forest improves generalization and reduces reliance on a small set of strong predictors.

Boosting is included because it follows a fundamentally different learning strategy. Unlike Bagging and Random Forest, which build trees independently, Boosting constructs trees sequentially, where each tree focuses on correcting the errors of the previous ones. This allows the model to pay more attention to difficult observations, such as misclassified defaulters, making it especially suitable for imbalanced classification problems.

By comparing these models, the study captures the trade-offs between interpretability, stability, and predictive performance, providing a comprehensive evaluation of different modeling approaches.

3. Methodology

3.1 Structure of the experiment

This study follows a structured experimental design to evaluate the effectiveness of different models in predicting credit card default. Before presenting any results, the methodology explains the evaluation criteria, hypotheses, variables, data characteristics, experimental setup, and comparison strategy.

Variables

The dependent variable in this study is the credit default status, which indicates whether a customer defaults on their credit card payment in the following month. This variable is binary, taking values of default and non-default.

The selection of independent variables is guided by financial domain knowledge and prior understanding of credit risk behavior, rather than model-driven significance. Each variable is included based on its expected relationship with default risk.

Repayment status variables, such as PAY_0, PAY_2, and PAY_3, represent the customer's recent payment behavior and are expected to be the strongest predictors of default. Customers with delayed or irregular payments are more likely to default in the future. Therefore, these variables directly capture financial reliability and are central to the model.

Billing amount variables, such as BILL_AMT1 through BILL_AMT6, reflect the customer's outstanding balance over time. Higher billing amounts may indicate increased financial burden, which can raise the probability of default. These variables help capture the scale and consistency of credit usage.

Payment amount variables, including PAY_AMT1 through PAY_AMT6, measure how much a customer pays toward their balance. Lower payment amounts relative to billing amounts may indicate financial stress or inability to repay, making these variables important indicators of default risk.

The credit limit variable, LIMIT_BAL, represents the maximum amount of credit extended to a customer. A higher credit limit may indicate financial trustworthiness, but it can also expose customers to higher debt levels. Its inclusion allows the model to capture both opportunity and risk associated with credit access.

Demographic variables such as AGE, EDUCATION, MARRIAGE, and SEX are included to examine whether personal characteristics contribute to default behavior. Age may reflect financial maturity and

stability, while education and marital status may correlate with income stability and financial responsibility. Although these variables are expected to have weaker predictive power compared to financial behavior variables, they are included to provide a comprehensive view of customer characteristics.

This variable selection approach ensures that all predictors are theoretically justified before model estimation, avoiding data-driven bias and improving the generalizability of the results.

Data Preparation and Characteristics

The dataset (Shashank Necrothapa. (n.d.). *Ames housing dataset* [Data set]. Kaggle.) was cleaned and prepared before modeling. The ID variable was removed because it does not provide predictive value. The target variable was converted into a categorical factor with two levels representing default and non-default cases. Categorical predictors such as education, marital status, and gender were encoded as factors, and rare categories were grouped into an "Other" category to improve stability.

The dataset was split into training and testing sets using stratified sampling, ensuring that the proportion of defaulters and non-defaulters remained consistent across both sets. This is particularly important because the dataset is imbalanced, with a higher number of non-default cases.

The dataset is appropriate for this study because it contains detailed financial and behavioral information relevant to credit risk. However, one limitation is the imbalance in the target variable, which may bias models toward the majority class. Another limitation is that the dataset primarily contains historical financial variables and does not include external factors such as macroeconomic conditions.

Evaluation Criteria

Due to class imbalance, accuracy alone is not sufficient to evaluate model performance. A model can achieve high accuracy by predicting the majority class while failing to identify defaulters.

For this reason, recall is treated as the most important metric, as it measures the proportion of actual defaulters correctly identified. In a financial context, missing a defaulter can lead to direct financial loss. The F1 score is used as a complementary metric because it balances recall and precision, ensuring that the model does not generate excessive false positives. Precision is also considered to evaluate the reliability of predicted default cases.

Hypotheses

This study is based on the hypothesis that tree-based ensemble methods will outperform the Lasso Logistic Regression baseline in detecting credit card defaulters. It is also expected that Boosting will achieve the best overall performance due to its sequential learning mechanism, which allows it to focus on difficult observations. Additionally, it is hypothesized that repayment history variables will be the most important predictors across all models.

Experimental Design

The experimental setup involves training multiple models and evaluating their performance on a held-out test dataset. Lasso Logistic Regression is used as a baseline model for comparison. Tree-based models, including Bagging, Random Forest, and Boosting, are trained with different parameter settings to examine how tuning affects performance.

For Bagging, the number of trees is varied to analyze its impact on model stability. For Random Forest, the number of predictors considered at each split is adjusted to study the trade-off between randomness and predictive strength. For Boosting, the number of trees, tree depth, and shrinkage parameter are varied to identify the optimal configuration.

Presentation and Comparison Strategy

The results are presented using confusion matrices and performance metrics tables to provide a clear comparison between models. Each model is evaluated based on accuracy, recall, precision, and F1 score. Comparison tables are used to identify the best-performing configurations within each method.

Variable importance plots are also used to analyze how each model prioritizes predictors. These plots help interpret model behavior and assess whether different methods agree on the most important variables.

The final comparison is conducted between the best-performing tree-based model and the Lasso baseline to determine the most suitable model for the problem.

3.2 Result

The quantitative results of all models are presented through structured tables and variable importance figures included in this report. Table 3.1 summarizes the performance of the baseline Lasso model and the selected tree-based models, while Tables 3.2, 3.3, and 3.4 present detailed experimental results for Bagging, Random Forest, and Boosting, respectively. Figures 3.1, 3.2, and 3.3 illustrate the variable importance for each tree-based method.

Table 3.1 shows that the Lasso Logistic Regression model achieves an accuracy of 0.8108, which is comparable to the tree-based models. However, its recall is only 0.2422, indicating that it correctly identifies less than one-quarter of actual defaulters. Although the precision is relatively high at 0.7120, this reflects a conservative prediction behavior where the model predicts fewer default cases. As a result, the F1 score is low at 0.3615, confirming that the model does not provide a balanced performance. This result suggests that the baseline model is not suitable for detecting defaulters despite its acceptable accuracy.

Model	**Accuracy**	**Recall**	**Precision**	**F1**
Lasso Logistic Regression	0.8108	0.2422	0.712	0.3615
Bagging (500 trees)	0.8142	0.3844	0.6312	0.4778
Random Forest (mtry =p/2)	0.816	0.3824	0.6406	0.4789
500 trees, depth 3, shrinkage 0.02	0.8202	0.3749	0.6661	0.4797

(Table 3.1: Lasso Logistic Regression Performance vs Selected tree-based models' performance)

The results for the Bagging model are reported in Table 3.2. The table shows that accuracy remains stable across different numbers of trees, ranging from 0.8124 to 0.8142. The highest accuracy is achieved at 500 trees. The recall values fluctuate slightly, with the highest recall of 0.3844 also observed at 500 trees. The F1 score follows a similar trend, reaching its maximum value of 0.4778 at this configuration. The small variation across configurations indicates that the model stabilizes after approximately 400 to 500 trees and further increases in the number of trees do not produce meaningful improvements.

Trees	Accuracy	Recall	Precision	F1
100	0.8124	0.3794	0.625	0.4722
300	0.8133	0.3784	0.6296	0.4727
400	0.8139	0.3804	0.6314	0.4748
500	0.8142	0.3844	0.6312	0.4778
600	0.8135	0.3824	0.6289	0.4756

(Table 3.2: Bagging Performance Across Different mtry Values)

Table 3.3 presents the results for the Random Forest model across different values of mtry. The best-performing configuration, where mtry is equal to half of the predictors, achieves an accuracy of 0.8160, recall of 0.3824, precision of 0.6406, and F1 score of 0.4789. Compared to Bagging, Random Forest maintains a similar recall but improves precision. This leads to a slightly higher F1 score, indicating a better balance between detecting defaulters and avoiding false positives. The variation across mtry values is relatively small, suggesting that the model is stable with respect to this parameter.

Mtry_Setting	Mtry_Value	Accuracy	Recall	Precision	F1
mtry = 2	2	0.8152	0.3513	0.6527	0.4567
mtry = sqrt(p)	4	0.8194	0.3754	0.6616	0.479
mtry = p/3	7	0.8179	0.3779	0.6522	0.4785
mtry = p/2	11	0.816	0.3824	0.6406	0.4789
mtry = 2	2	0.8152	0.3513	0.6527	0.4567

(Table 3.3: Random Forest Performance Across Different mtry Values)

The Boosting results are shown in Table 3.4. The model demonstrates sensitivity to parameter settings, particularly tree depth and shrinkage. The best-performing configuration, with 500 trees, depth 3, and shrinkage 0.02, achieves the highest accuracy of 0.8202 among all models. It also produces a recall of 0.3749, precision of 0.6661, and the highest F1 score of 0.4797. Although its recall is slightly lower than that of Bagging and Random Forest, its higher precision results in the best overall balance. The differences in performance between Boosting and Random Forest are small but consistent, indicating a practical advantage.

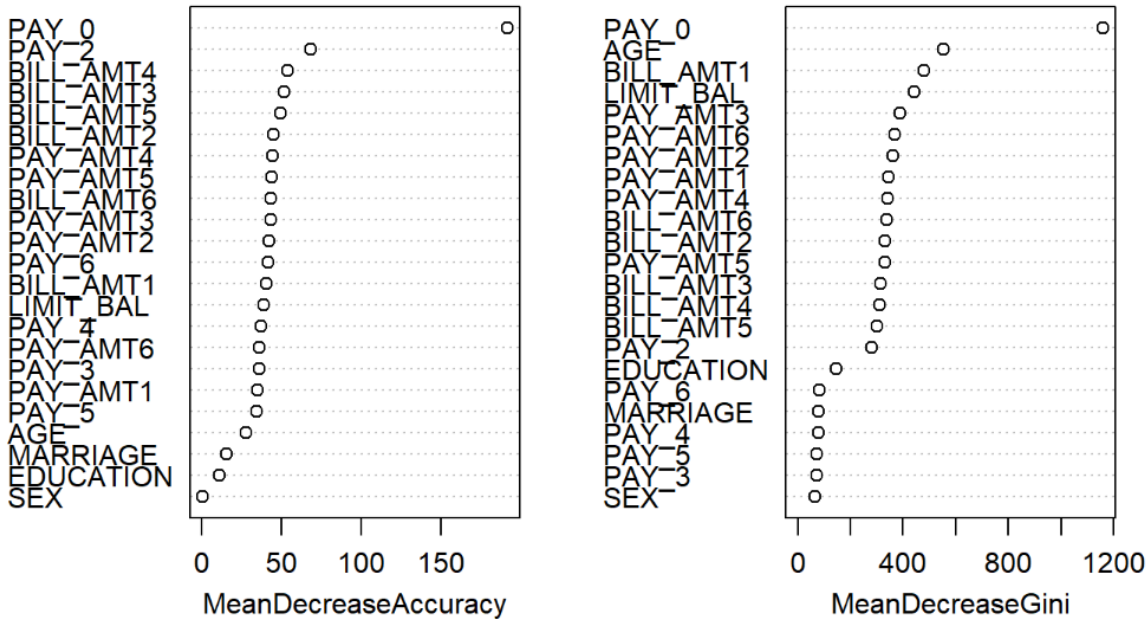
Settings	Accuracy	Recall	Precision	F1
500 trees, depth 3, shrinkage 0.01	0.8213	0.3709	0.6746	0.4786
1000 trees, depth 3, shrinkage 0.01	0.8192	0.3739	0.6613	0.4777
500 trees, depth 1, shrinkage 0.01	0.8155	0.3126	0.6805	0.4284
500 trees, depth 5, shrinkage 0.01	0.8203	0.3683	0.6706	0.4755
500 trees, depth 3, shrinkage 0.005	0.8213	0.3623	0.6802	0.4728
500 trees, depth 3, shrinkage 0.02	0.8202	0.3749	0.6661	0.4797

(Table 3.4: Boosting Performance Across Different Parameter Settings)

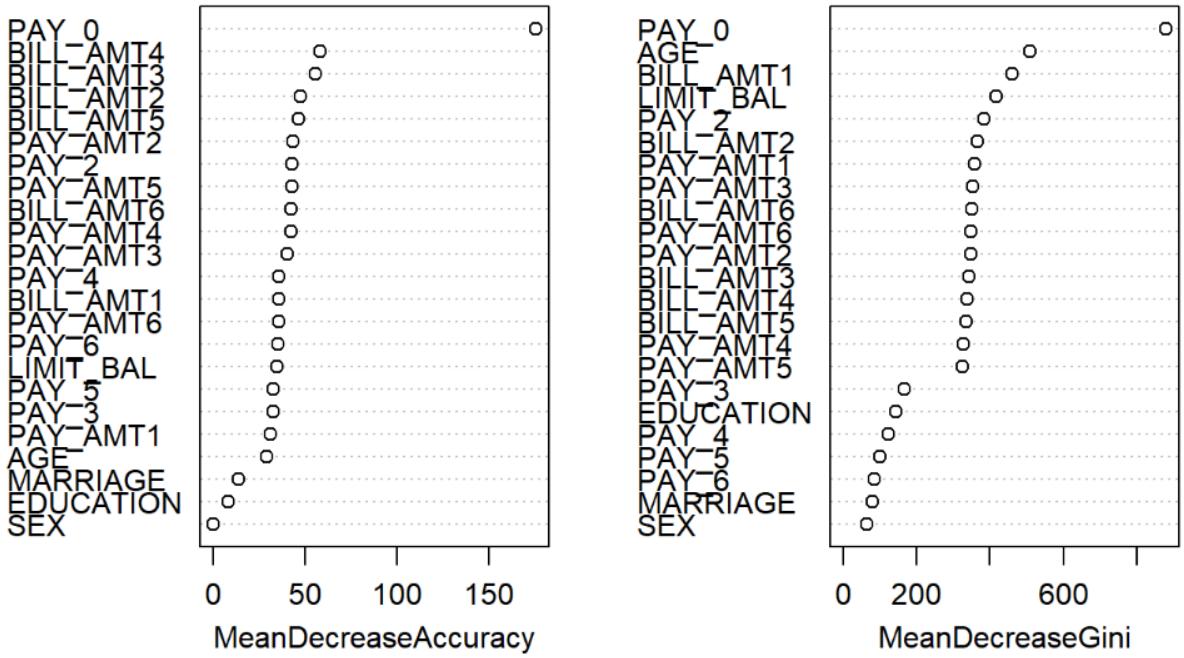
The variable importance plots further support these findings. Table 3.5 presents the variables retained by Lasso base line model. Figure 3.1, which presents the Bagging importance plot, shows that PAY_0 is the most influential variable, followed by PAY_2 and several BILL_AMT variables. Figure 3.2 shows a similar pattern for the Random Forest model, where PAY_0 again dominates, and repayment-related variables consistently rank higher than demographic variables. Figure 3.3, which presents the Boosting importance plot, shows an even stronger concentration of importance on PAY-related variables, particularly PAY_0, PAY_2, and PAY_3. Across all figures, demographic variables such as SEX, EDUCATION, and MARRIAGE consistently appear at the lower end of the importance rankings.

Variable	**Coefficient**
EDUCATIONOther	-0.9715053
PAY_0	0.5754504
MARRIAGESingle	-0.1835572
PAY_2	0.1032092
SEXFemale	-0.0986683
EDUCATIONHigh_School	-0.0956657
MARRIAGEOther	-0.0815684
EDUCATIONUniversity	-0.0620641
PAY_3	0.0555545
PAY_4	0.0378941
PAY_5	0.0336253
AGE	0.005804
PAY_AMT1	-0.0000095
PAY_AMT2	-0.0000081
PAY_AMT3	-0.0000035
PAY_AMT5	-0.0000034
BILL_AMT1	-0.0000027
PAY_AMT4	-0.0000018
PAY_AMT6	-0.0000013
BILL_AMT5	0.000001
LIMIT_BAL	-0.0000008
BILL_AMT3	0.0000003

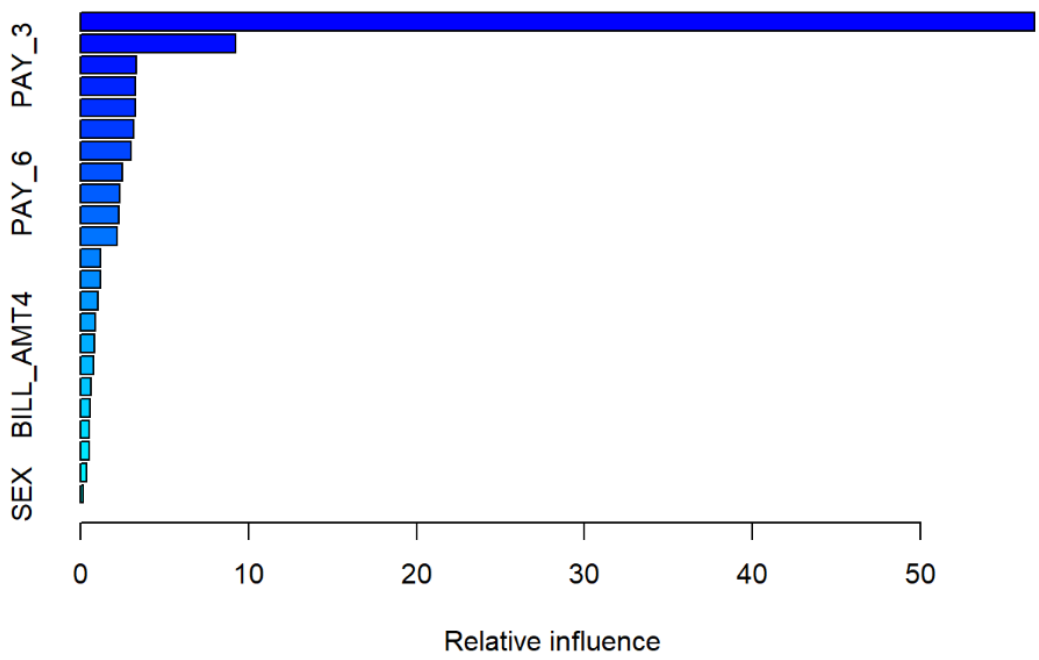
(Table 3.5: Variables Retained by Lasso)



(Figure 3.1: Variable Importance (Bagging Model-500 trees))



(Figure 3.2: Variable importance Random Forest Model (mtry = p/2))



(Figure 3.3: Variable Importance (Boosting))

Overall, Table 3.1 demonstrates that all tree-based models significantly outperform the Lasso baseline in terms of recall and F1 score. The increase in recall from 0.2422 to approximately 0.38 represents a substantial improvement in identifying defaulters. Tables 3.2, 3.3, and 3.4 further show that performance differences among the tree-based models are relatively small, indicating that model performance has reached a stable range. However, Boosting provides the highest F1 score, while Random Forest offers slightly better stability. These results confirm that ensemble tree-based methods are more effective than linear models for this classification problem.

Unlike traditional statistical models, where significance is evaluated using p-values, machine learning models are assessed based on predictive performance and stability across different parameter settings. In this study, the consistency of performance metrics such as recall and F1 score across multiple configurations indicates that the observed differences are robust rather than driven by random variation. Therefore, model comparison is based on practical significance in predictive performance rather than formal hypothesis testing.

3.3 Discussion

The results of this study provide strong evidence in support of the initial hypotheses. It was hypothesized that tree-based ensemble methods would outperform the Lasso Logistic Regression baseline in detecting credit card defaulters, that Boosting would achieve the best overall performance, and that repayment-related variables would emerge as the most important predictors. The findings largely confirm these expectations, while also revealing important nuances regarding model behavior, trade-offs, and limitations.

The most significant outcome of the analysis is the clear performance gap between the baseline model and the tree-based methods. The Lasso model, despite achieving an accuracy of 0.8108, performs poorly in identifying defaulters, with a recall of only 0.2422. This confirms that accuracy alone is not a sufficient measure in imbalanced classification problems. The model's high precision indicates that it is conservative in predicting defaults, but this comes at the cost of missing a substantial number of actual defaulters. This behavior can be explained by the linear nature of Lasso Logistic Regression, which assumes a linear relationship between predictors and the log-odds of the outcome. Such an assumption is overly restrictive for financial behavior data, where nonlinear relationships and interactions are common.

In contrast, all three tree-based methods significantly improve recall, increasing it to approximately 0.38. This represents a meaningful improvement in the model's ability to detect defaulters. The improvement is not marginal but substantial in practical terms, as it means that a much larger proportion of high-risk customers can be identified. This confirms the hypothesis that nonlinear ensemble methods are better suited for this type of problem, as they are capable of capturing complex relationships that linear models cannot.

Among the tree-based models, Boosting achieves the highest F1 score of 0.4797, indicating the best balance between recall and precision. Although its recall is slightly lower than that of Bagging and Random Forest, the higher precision compensates for this difference, resulting in a superior overall performance. This outcome aligns with the theoretical properties of Boosting, which focuses on correcting previous errors through sequential learning. By iteratively adjusting the model to focus on misclassified observations, Boosting is able to refine decision boundaries more effectively than methods that rely on independent tree construction.

Random Forest demonstrates strong and stable performance across different parameter settings, with an F1 score of 0.4789 and recall of 0.3824. Its ability to maintain performance consistency can be attributed to the combination of bootstrap sampling and feature randomness, which reduces overfitting and improves generalization. Bagging, while slightly less effective in terms of F1 score, achieves the highest recall among the tree-based models. This suggests that Bagging is particularly effective at capturing the minority class, although it does so with slightly lower precision.

The relatively small differences in performance among the three tree-based methods suggest that all of them have reached a similar level of predictive capability given the available data. The variation in F1 scores across these models is minimal, indicating that further improvements may require either additional features or more advanced modeling techniques rather than simple parameter tuning. However, the

consistent superiority of all tree-based models over the baseline confirms that model choice plays a critical role in handling imbalanced classification problems.

The variable importance analysis provides additional insight into why these models perform well. All models consistently identify repayment history variables, particularly PAY_0, as the most important predictors. This supports the hypothesis that financial behavior variables dominate credit risk prediction. The strong influence of these variables suggests that recent repayment behavior is a key indicator of future default risk. In contrast, demographic variables such as gender, education, and marital status are consistently ranked as least important across all models. This indicates that personal characteristics contribute little to predictive performance compared to behavioral financial variables.

The results also highlight important trade-offs between model complexity, interpretability, and stability. While Boosting achieves the best overall performance, it is more complex and sensitive to parameter tuning. Random Forest provides a more stable alternative with comparable performance, making it a practical choice in scenarios where robustness is prioritized. Bagging offers simplicity and consistency but does not achieve the same level of performance as the other methods.

Despite the strong results, the study has limitations. The dataset is imbalanced, which continues to affect the ability of all models to fully capture the minority class. Although recall improves significantly with tree-based methods, it remains below 0.40, indicating that a portion of defaulters is still not detected. Additionally, the dataset does not include external factors such as economic conditions or customer behavioral trends over longer periods, which could further enhance predictive performance.

In summary, the results demonstrate that the hypotheses are largely supported. Tree-based ensemble methods outperform the baseline model, Boosting provides the best overall performance, and repayment history variables are the dominant predictors. At the same time, the findings reveal that improvements beyond the current level will likely require more advanced techniques or richer data rather than further tuning of existing models.

4. Limitations and Future Work

Although the analysis provides meaningful insights into credit card default prediction, several limitations remain in both the data and modeling approach. Addressing these limitations would likely improve the robustness and predictive performance of the models.

One of the primary limitations of this study is the class imbalance present in the dataset. The number of non-default cases significantly exceeds the number of default cases, which biases the models toward the majority class. Although recall was used as a key evaluation metric to address this issue, all models still achieved recall values below 0.40, indicating that a substantial proportion of defaulters remains undetected. This limitation could be addressed by applying resampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE) or by introducing class-weighted learning, where misclassification of the minority class is penalized more heavily. These approaches would force the model to pay greater attention to default cases and could improve recall.

Another limitation lies in the feature set used in the analysis. The dataset primarily includes historical repayment behavior, billing amounts, and basic demographic variables, but does not incorporate external or behavioral factors such as income level, employment stability, or macroeconomic conditions. These missing variables may contain important signals related to default risk. Future work could involve integrating additional data sources or constructing new features that capture temporal trends, such as changes in payment behavior over time. This would allow the models to better capture dynamic financial behavior and potentially improve predictive accuracy.

The modeling approach itself also introduces limitations. While tree-based ensemble methods such as Bagging, Random Forest, and Boosting capture nonlinear relationships effectively, they lack interpretability compared to simpler models like Lasso Logistic Regression. This can be problematic in financial applications where transparency and explainability are important for decision-making. To address this limitation, future work could incorporate model-agnostic interpretability techniques such as SHAP values or partial dependence plots. These methods would provide more detailed explanations of how individual predictors influence model predictions.

Another limitation is the relatively small performance gap among the tree-based models. The differences in F1 score between Bagging, Random Forest, and Boosting are minimal, suggesting that model performance has reached a plateau given the current data and feature set. This indicates that further improvements may not be achieved through parameter tuning alone. Future work could explore more advanced models such as gradient boosting frameworks like XGBoost or LightGBM, which offer additional regularization and optimization capabilities. These models may provide incremental improvements in performance.

The evaluation strategy used in this study also has limitations. The models were evaluated using a single train-test split, which may introduce variability in performance estimates depending on how the data is partitioned. A more robust approach would involve using cross-validation, such as k-fold cross-validation, to obtain more reliable estimates of model performance. This would reduce the risk of overfitting to a particular split and provide a more stable assessment of model generalization.

Finally, the decision threshold used for classification was fixed and not optimized. All models used a default threshold to convert predicted probabilities into class labels. However, in imbalanced classification problems, adjusting the decision threshold can significantly improve recall without excessively reducing precision. Future work could involve optimizing the threshold based on business objectives, such as minimizing financial loss or maximizing detection of high-risk customers.

In summary, while the current analysis demonstrates the effectiveness of tree-based models for credit default prediction, several improvements are possible. Addressing class imbalance, enriching the feature set, enhancing interpretability, adopting more advanced models, improving evaluation methods, and optimizing decision thresholds would all contribute to a more robust and practically useful predictive framework.

4. Conclusion

This study examined the effectiveness of different modeling approaches for predicting credit card default, with a particular focus on comparing a baseline Lasso Logistic Regression model with tree-based ensemble methods, including Bagging, Random Forest, and Boosting. The results demonstrate that model choice plays a critical role in handling imbalanced classification problems and in accurately identifying high-risk customers.

A key finding of this analysis is that while the Lasso model achieves a reasonable level of overall accuracy, it performs poorly in detecting defaulters, as reflected by its low recall. This highlights the limitation of relying on accuracy alone in imbalanced datasets, where correct classification of the majority class can mask poor performance on the minority class. In contrast, all three tree-based methods significantly improve recall, indicating a stronger ability to identify customers who are likely to default.

Among the tree-based models, Boosting emerges as the best-performing method in terms of overall balance between recall and precision, achieving the highest F1 score. Although Bagging achieves slightly higher recall, and Random Forest provides strong stability across parameter settings, Boosting offers the most effective trade-off between detecting defaulters and minimizing false positives. This result demonstrates the advantage of sequential learning, where the model iteratively improves by focusing on previously misclassified observations.

Another important contribution of this study is the identification of key predictors of credit default. Across all models, repayment history variables, particularly recent payment status indicators, are consistently identified as the most influential factors. In contrast, demographic variables such as gender, education, and marital status contribute minimally to prediction accuracy. This finding suggests that credit default behavior is primarily driven by financial behavior rather than personal characteristics.

The results of this study have practical implications for financial institutions. The use of advanced ensemble methods, particularly Boosting, can significantly improve the identification of high-risk customers, thereby reducing potential financial losses. At the same time, the findings highlight the importance of selecting appropriate evaluation metrics, as relying solely on accuracy can lead to misleading conclusions in imbalanced classification problems.

In conclusion, this study demonstrates that tree-based ensemble methods provide a more effective and reliable approach for credit card default prediction than traditional linear models. The findings emphasize the importance of model selection, evaluation strategy, and feature relevance in developing robust predictive systems. Future work can build on these results by incorporating additional data sources, addressing class imbalance more effectively, and exploring more advanced modeling techniques to further enhance predictive performance.

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